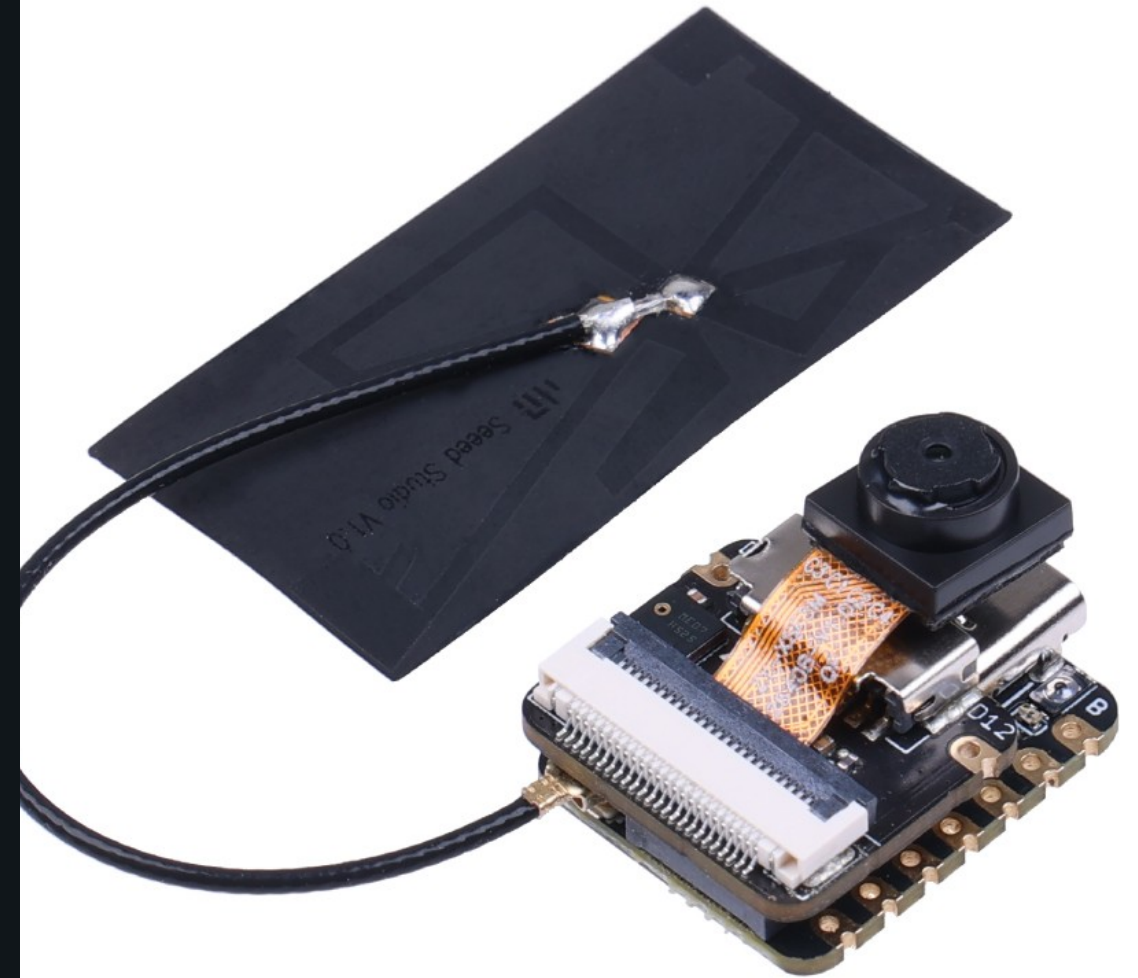


ECE 116

Advanced Hardware & Software Design

Intelligent embedded systems | Fall 2026



Instructional Team

Course staff, office hours, room, and meeting time are published on Canvas.



Professor Edward Wang

ejaywang@ucsd.edu



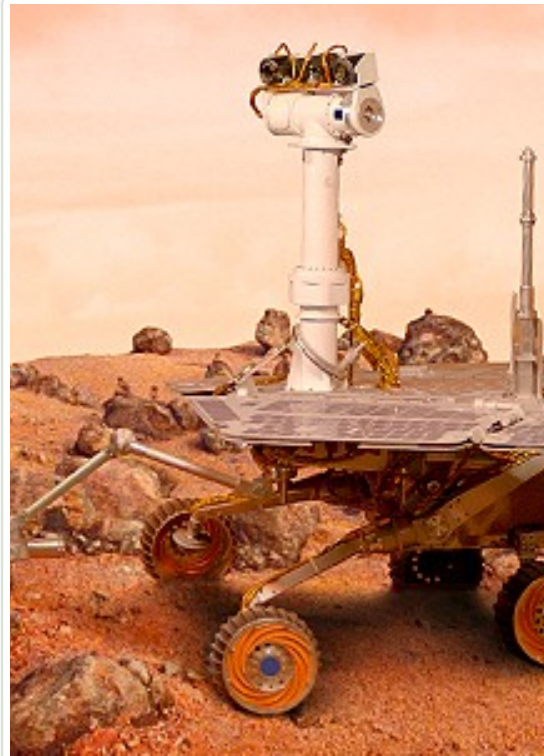
Pamely Zantou

pzantou@ucsd.edu

Intelligent systems live where software meets the physical world



health



exploration



assistive care



automation

Different applications. The same need to connect computation to real conditions and consequences.

A useful device must observe, decide, and act

observe a request
or condition

maintain state
and make a decision

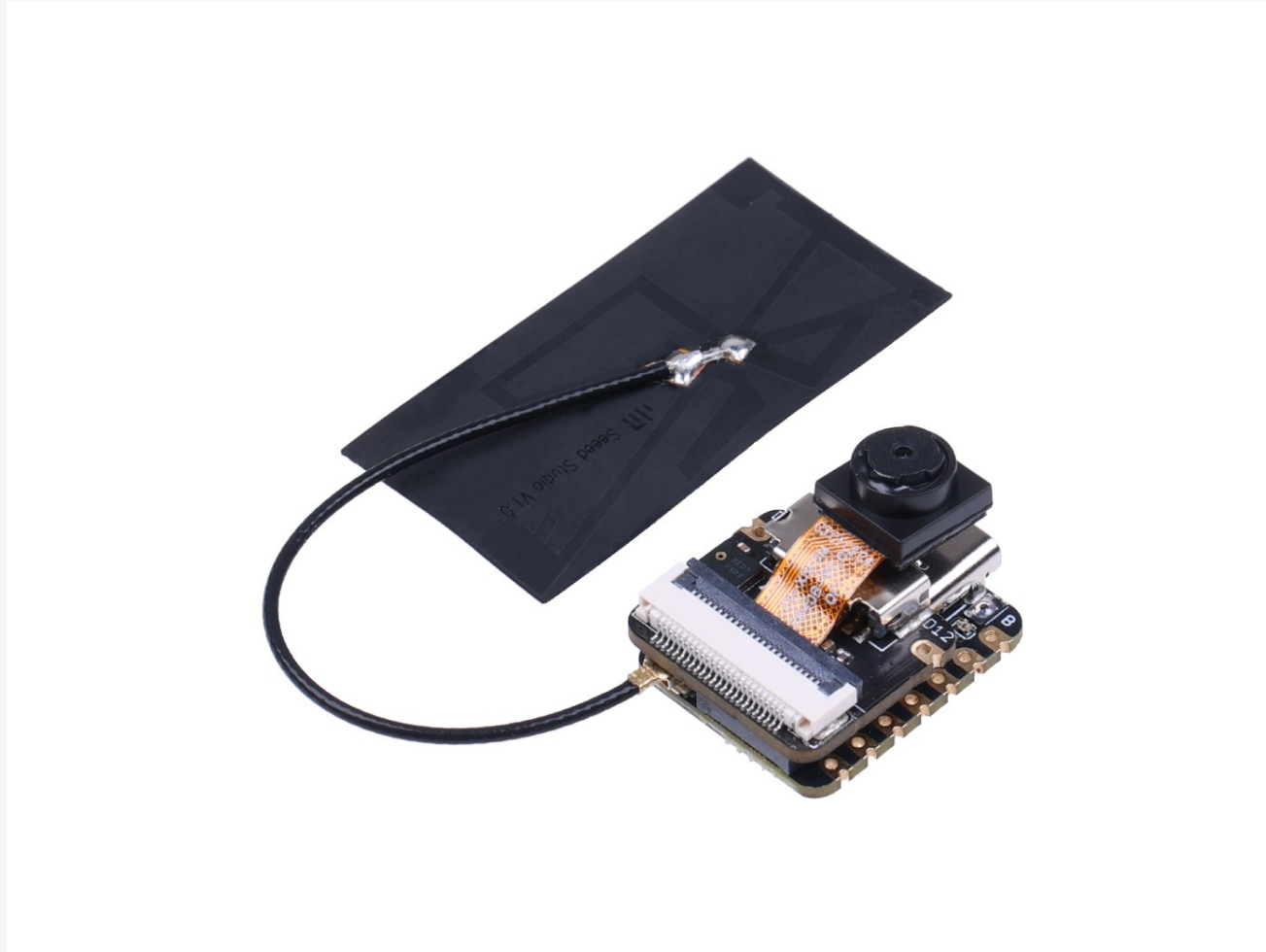


act in the
physical world

make the result
observable

The difficult bugs often live in the interfaces between these responsibilities.

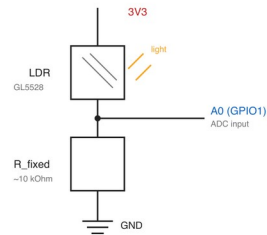
XIAO ESP32-S3 Sense



- camera
- microphone
- GPIO, ADC, and buses
- USB power, upload, and serial

Assignments

GL5528 LDR voltage divider



Brighter light lowers the LDR resistance, so the A0 node rises with light and falls when covered. Confirm the direction and dark level on YOUR board.

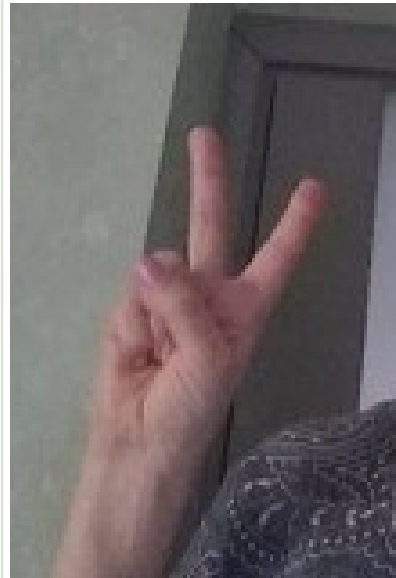
A1

**Microcontroller &
Finite State Machine**



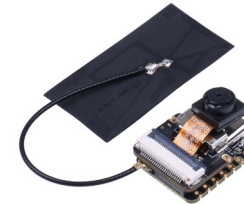
A2

**Evaluating a Provided
Voice Model**



A3

**Training a Vision
Model**



A4

**Vision Deployment
Under Constraints**

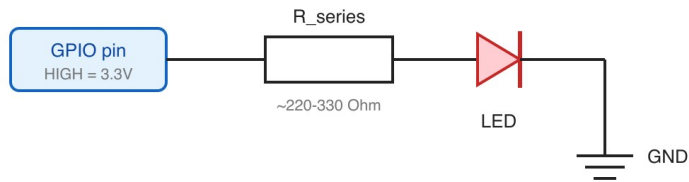


A5

**Cloud Connectivity
and Security**

Assignments

Mode LED with current-limiting resistor (one per LED)

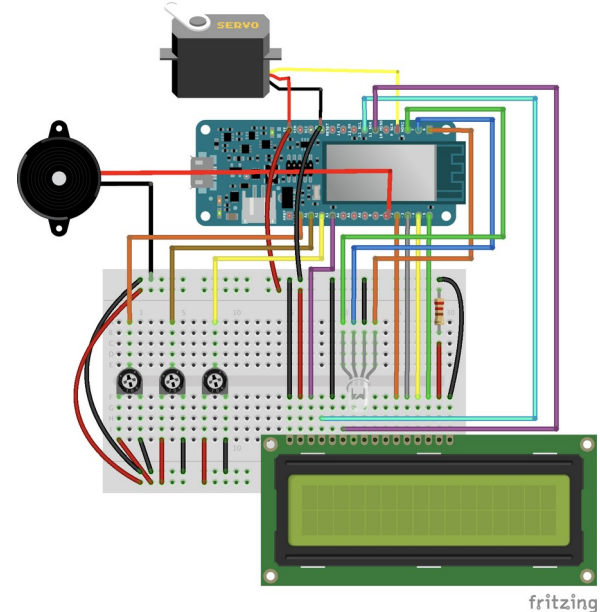


Sizing: $R = (V_{\text{supply}} - V_f) / I_{\text{target}}$

Different colors have different forward voltage V_f :

red ~1.8V, green ~2.1V, blue/white ~3.0V. Pick R per color; ~330 Ohm is a safe default.

In-class labs: controlled mechanism



Challenge: new system context

Illustrations show the lab/challenge pattern. Use each handout for the required hardware.

Class Schedule

Both days: 9:30 a.m.–12:20 p.m. Pacific

Tuesday

LECTURE
(Zoom)

Conceptual overviews

Thursday

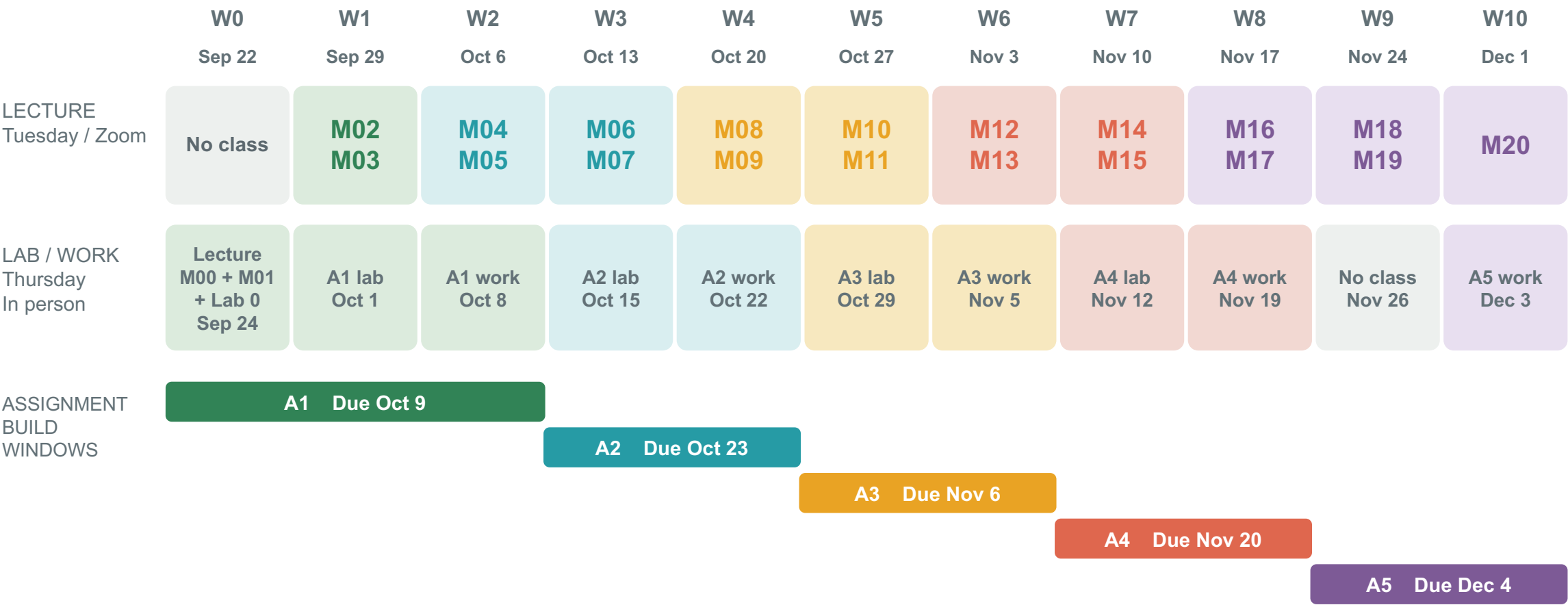
LAB OR OPEN WORK
Jacobs Hall, B538

Bring the XIAO, laptop, a known-good data cable, and the components required for hardware work.

First meeting: Thu Sep 24, in person. M00 + M01 and Lab 0.

Course Cadence

Lectures introduce the next topic during the current assignment's due week.

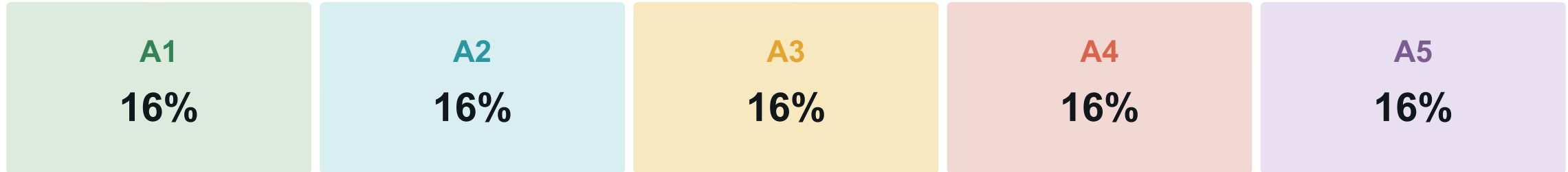


W0: M00 + M01 and Lab 0 meet Thursday in person. All deadlines: Friday, 7 p.m. Pacific.

Course Grade Components

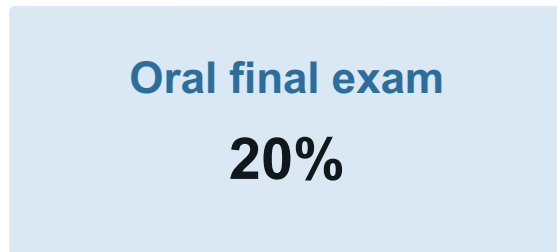
Assignments

80%



Individual oral examination

20%



30-minute appointment during finals week

Oral Final Exam



In-Person: 30-minute appointment during finals week

Finding course materials in Canvas

I need to...	Open...
Get set up for the course	<u>Home</u> → <u>Start Here</u>
Find this week's slides and preparation	<u>Modules</u> → <u>Current week</u>
Check the quarter schedule or policies	<u>Syllabus</u>
Download an assignment, read its rubric, or submit	<u>Assignments</u> → <u>A1–A5</u>
See course updates	<u>Announcements</u>
Find grades and feedback	<u>Grades</u>

Material links open on GitHub. Download the assignment ZIP and submit your completed work in Canvas.

Late Policy

FRI 7:00 PM



posted deadline

FRI 11:59 PM



automatic grace ends

MON 7:00 PM



submission closes

**No late deduction
Through Fri 11:59 p.m.**

**Deduct 25% of earned score
Sat midnight–Mon 7 p.m.**

**Submission closed
After Mon 7 p.m.**

All times Pacific. Approved individual accommodations and extensions apply.

Collaboration Policy

WORK TOGETHER

discuss concepts
compare test strategies
interpret error messages
work near classmates

SUBMIT YOUR OWN

design + code
data + measurements
evidence + figures
written + oral reasoning

Do not exchange ready-to-submit work or use another student's measurements as your own. University academic integrity policies apply.

AI Use in This Course

Allowed

- ✓ Ask for an explanation of a concept
- ✓ Debug a specific error or compare approaches
- ✓ Improve clarity after you understand the work
- ✓ Use feedback to revise your own design

Not allowed

- ✗ Submit generated work you cannot explain
- ✗ Fabricate measurements, screenshots, or evidence
- ✗ Use AI during an oral assessment
- ✗ Bypass collaboration or citation requirements

Disclose substantive AI use in your README: tool, help received, and what you verified.

Lab 0: Kit setup and bring-up

Collect your kit. Solder headers with staff and inspect before powering the board.



compile

upload

serial

LED

Bring your laptop and a USB data cable. Lab 0 has no graded submission.